

# BrickAtlas: An Auditable Casebank for Multi-Task Visual and Symbolic Brick-Structure Reasoning

Development Manuscript

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September 12, 2026

## Abstract

*Brick-structure reasoning combines recognition, spatial relations, reconstruction, generation, editing, and executable planning. These tasks require different inputs and notions of correctness: a plausible hidden layout need not match one reference, while a submitted assembly order must actually execute. We present the current development release of BrickAtlas, a versioned casebank with 5,120 distinct procedural geometries, 117,760 object-derived task-condition cases, and 150 separate known-part questions. Each case includes a machine-readable ground-truth record, an independently implemented integer-cell geometry certificate, applicable metrics, and replayable evaluation evidence. We separate ordinary RGB, layer disclosure, and privileged symbolic references, and distinguish model submissions from harness operations and evaluator diagnostics. On the complete casebank, a height-sorted planner solves all 10,240 assembly/disassembly cases, whereas greedy assembly solves only 2,364 of 5,120. Thus the present planning domain diagnoses constraint following, not difficult combinatorial search. A new two-model v2 study contains 674 one-shot calls and 56 controls on 24 shared held-out objects and 25 catalog questions. Both models fail exact ordinary reconstruction; privileged symbolic references solve all eight controlled reconstructions. All 117,910 v2 oracle answers pass, and 138,240 independently computed metric comparisons agree. The release is an executable research resource, not yet a confirmatory benchmark study: semantic diversity, independent human review, and broader empirical coverage remain limited.*

## 1. Introduction

Building a structure from parts requires more than recognizing its final appearance. A system may identify a brick yet misunderstand a connection, produce a plausible exterior with a different hidden arrangement, or specify a legal final assembly in an impossible construction order. Conversely,

failure to recover hidden geometry from an opaque image does not necessarily imply a reasoning failure. A benchmark must make these distinctions explicit before scores can support claims about model ability.

Prior work establishes strong foundations. BrickGPT generates buildable structures from text using autoregressive prediction with validity checking and physics-aware rollback [4]. BrickNet expands the data and connection vocabulary through human-designed LDraw objects and graph-backed assembly programs [2]. Break and Make evaluates interactive understanding by inspecting, disassembling, and rebuilding objects in LTRON [6]. BrickAtlas does not claim to originate generation, connection reasoning, or disassembly-and-reconstruction. Its present focus is an explicit evaluation contract connecting several task families to the same source objects and making input privileges, alternative answers, and execution evidence inspectable.

Three practical problems motivate this release. First, dataset size is easily overstated by counting colors, views, and task variants as independent objects. Second, “ground truth” may refer either to one valid witness or to the only acceptable answer. Conflating them penalizes legitimate alternatives. Third, a model’s output can be improved or re-ordered by the surrounding harness without that intervention being visible in the reported result. A visualization of a final structure being unfolded piece by piece is not evidence that the model planned its assembly.

We address these problems with a procedural casebank, typed evaluation, and auditable artifacts. The scope is deliberately digital: orthogonal rectangular bricks and plates, nominal stud contact, and vertical insertion/removal. We do not use a robot, manipulator, or vision–language–action controller. We do not equate nominal support with physical stability.

The current contribution package is:

1. A versioned resource with 5,120 geometrically distinct objects from eight construction policies, preserving source-group membership across all derived tasks and input conditions.
2. Per-case ground truth that records answer semantics,

structural targets, protected regions, nominal connection ports, executable witnesses, and metric-specific oracle evaluations.

3. A complete deterministic baseline and replay audit, including counterexamples that exposed ambiguous repair labels and metadata leakage during development.

These are implemented capabilities, not claims of state-of-the-art performance or submission readiness. New v2 model evidence and earlier pilots are reported separately rather than pooled across dataset versions.

## 2. Related Work and Positioning

**Generation and structural representations.** BrickGPT [4], previously released under the LegoGPT name, couples a language model with external checks to generate stable brick structures from text. BrickNet [2] collects over 100,000 human-designed objects and scenes and models connectivity through graph-backed programs. These works cover substantially richer semantic or geometric settings than the current BrickAtlas grid domain. Our relative-coordinate encoding, inherited from an earlier release, is not a replacement for BrickNet’s typed connector language. The present casebank asks a different engineering question: how should multiple tasks on shared objects be labeled, scored, and traced without silently changing the information available to a model?

**Interactive assembly.** Break and Make [6] already studies structural understanding through disassembly and reconstruction. BrickAtlas’s earlier CARE-mini track also uses an interactive action loop, but the v2 casebank evaluated here is primarily a fixed-input submission protocol. A static structure answer, a submitted plan, a fixed validation-feedback run, and autonomous tool use are separate tracks. We do not merge their scores into a single agent leaderboard.

**Training baselines.** We use a public Qwen3-0.6B checkpoint [5] with standard LoRA [1] for an earlier text/structure pilot. This is an implementation baseline, not a new model architecture. The experiment does not train a vision encoder, does not use v2 data, and does not establish compute-matched multi-task transfer.

## 3. Casebank Construction

### 3.1. Geometric Domain

A part is represented as

$$p = (\text{id}, t, c, x, y, z, r), \quad (1)$$

where  $t$  is one of 25 catalog types,  $c$  is one of six colors,  $(x, y, z)$  is the minimum body corner, and  $r \in \{0, 1, 2, 3\}$  is

| Split      | Source objects | Task-condition cases |
|------------|----------------|----------------------|
| train      | 2,690          | 61,870               |
| validation | 392            | 9,016                |
| test_id    | 758            | 17,584               |
| test_ood   | 1,280          | 29,440               |
| Total      | 5,120          | 117,910              |

Table 1. V2 counts. The ID case count additionally includes 150 isolated part/color questions, which are not counted among the 5,120 objects.

a quarter-turn yaw. Horizontal units are studs (8 mm), and vertical units are plate heights (3.2 mm). Coordinates are integers; the inherited validator limits part origins to  $[0, 24]^3$ , bodies to  $[0, 32]^3$ , and submissions to 64 parts. Generated v2 objects contain 9–47 parts.

Nominal validity requires no body overlap and at least one supporting stud below every elevated part. Planning additionally checks unobstructed vertical insertion or removal. All structures start with a common 4-by-8 plate; this improves controlled generation but restricts shape diversity. Types and colors are public rather than novel categories. Actual catalog coverage of assembled objects is smaller than the 25-type recognition catalog.

### 3.2. Policies, Groups, and Splits

We generate 640 accepted unique geometries per policy (Fig. 1). Tip walk attaches at the current tip with one supporting parent; low fan samples low parents; shallow terrace restricts attachment height more strongly; plate network uses thin parts; column field constructs independent vertical profiles on a common base; alternating beams switch beam orientation. Two-support bridge explicitly constructs paired supports before spanning them, while enclosed shell constructs a perimeter and roof around varying interiors. These are different construction procedures, not a claim of eight disjoint semantic distributions.

Geometry is canonicalized over color removal, translation, and four global yaw rotations. Exact duplicate geometries are rejected both within v2 and against the two earlier procedural releases. Mirroring is not included in the equivalence group. All variants of an accepted object inherit one group and split.

Six policies use a deterministic geometry-hash partition into training, validation, and ID test objects. Bridge and shell policies are entirely held out. This is a *policy holdout*: it does not prove that primitive motifs or geometric support are absent from training. Unlike an earlier pilot whose differently named families sometimes shared sampling code, v2 declares the actual policy distinction and its limits.

Table 1 reports the realized partition. Piece-count difficulty bins contain 1,445 small ( $\leq 16$ ), 2,141 medium (17–

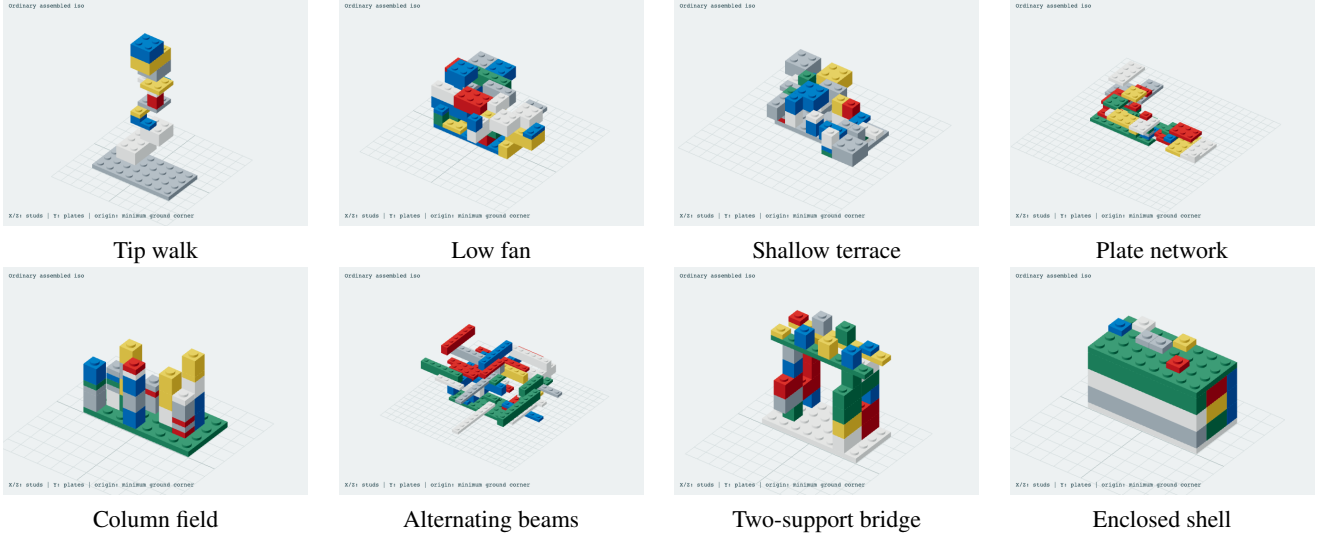


Figure 1. Rendered v2 examples, selected as the largest sample within each policy for a rendering stress test, not a random semantic showcase. Bridge and shell construction policies are reserved for the policy-held-out split. Shared grid primitives and the common root plate remain present across splits.

28), and 1,534 large ( $> 28$ ) structures. They are mechanical bins, not measured human difficulty.

We run a lossless length/prefix Jaccard join on canonical part-position sets at threshold 0.8, then cross-check it by brute force over all 8,305,836 cross-split pairs. No pair reaches the threshold. This is exhaustive only for the declared canonical token representation: canonical orientation can change under perturbations, and the check is not a best-aligned geometric distance, semantic leakage test, or pre-training contamination audit.

### 3.3. Task Expansion and Accounting

Each object produces 13 task variants and 23 task-condition cases: two relation queries (connected and separated), one reconstruction, one constrained generation, one suffix completion, three edits (rotation, recoloring, removal), two plans (assembly, disassembly), and three repairs (normal, color, shift). Reconstruction, completion, and each repair variant have three information conditions. Thus,

$$N_{\text{variant}} = 5,120 \times 13 + 150 = 66,710, \quad (2)$$

$$N_{\text{case}} = 5,120 \times 23 + 150 = 117,910. \quad (3)$$

The extra 150 cases cover 25 known catalog types in six colors. They form a separate recognition set rather than repeating the same isolated image once per assembly. Color instances share the catalog-type group when uncertainty is summarized.

## 4. Ground Truth and Information Boundaries

### 4.1. A Witness Is Not Always a Unique Answer

Every case stores a ground-truth object containing the public-input hash, source group and split, target and starting geometry, bill of materials (BOM), extents, connection edges, individual nominal stud mates, changed and protected IDs, yaw symmetry, assembly and disassembly witnesses, applicable metric names, and an evaluated oracle answer. The record explicitly declares whether the answer is exact or whether the oracle is only one member of an acceptable set.

For generation, the witness satisfies dimensions, zero anchoring, piece budget, color count, connectivity, and branching constraints; other satisfying layouts are accepted. For planning, any executable permutation is valid. Structure reconstruction ignores arbitrary instance IDs and symmetric yaw rotations. These equivalences are different and are not collapsed into one similarity score.

### 4.2. Three Information Conditions

**Ordinary RGB.** Four assembled views (isometric, top, front, side) and a BOM are provided for reconstruction, completion, and repair. Hidden layers are not revealed. The primary structural criterion accepts matching colored positive-axis first-hit surfaces plus BOM and nominal validity. It therefore does not force one unobservable interior arrangement. Preservation and localization gates remain mandatory where relevant.

**Layer disclosure.** An assembled view and separate top views for each occupied bottom-height layer reveal additional information. Unlike ordinary RGB, this condition requires the complete reference type/color/pose multiset, up to symmetry. It is not pooled with ordinary visual performance.

**Privileged symbolic reference.** The complete reference program is explicitly supplied. This diagnostic isolates failures that remain despite complete geometry information and requires full structural correctness. It is not a visual benchmark score. Other task families use their declared default inputs: symbolic structures for relations, edits, and plans; constraints for generation; isolated images for part recognition.

### 4.3. Independent Checks and Label Repairs

Generation uses the main axis-aligned geometry checker. A separately implemented integer-cell checker independently traverses occupied cells, support, collision, and insertion corridors. Connection stud totals must agree. Every ground-truth witness passes this cross-check and its task scorer. This independence is at the algorithm-implementation level; both checkers share the declared catalog and nominal geometric assumptions. It is not independent human validation.

A further independent implementation checks multiset, occupancy, colored surface, and edge metrics on 15,360 original/perturbed structures: 138,240 comparisons agree. An independently relaxed shortest-path algorithm agrees on all 10,240 relation queries. Nominal connector counts agree on all 5,120 structures, still only within the grid catalog. An additional 107,520 mutation/invariance checks cover missing, duplicate, unknown, and reversed plan actions; no-op edits; wrong repair localization; and instance-ID/yaw equivalence. These finite checks are not a proof over all possible malformed submissions.

Full-case baseline testing exposed a subtle repair ambiguity during pre-release development. A shifted brick could occupy the same pose as another brick of the same type and color. The reference image then could not identify which of the two current IDs should be labeled faulty. We exclude these indistinguishable shifted poses before generating the release. Thirty cases exposed the issue in an initial symbolic repair baseline. The final casebank and baseline were rebuilt and re-audited; pre-release results are not reported as final scores.

Fault type, changed IDs, target programs, and private render geometry are not carried in ordinary public input records. In particular, the normal/color/shift variant label is evaluator metadata, not a prompt field. The model exporter emits only a case ID, fixed system prompt, public task, and image paths. The full case index and ground-truth shards remain public research artifacts, so this separation

is a protocol boundary, not protection against a participant deliberately consulting the answers.

## 5. Evaluation

### 5.1. Structural and Task-Specific Metrics

Let  $T$  and  $A$  denote expected and submitted part multisets, and let  $m$  be the number of exact type/color/pose matches after accounting for yaw symmetry. Part precision, recall, and F1 are

$$P = \frac{m}{|A|}, \quad R = \frac{m}{|T|}, \quad F_1 = \frac{2m}{|A| + |T|}. \quad (4)$$

The implementation defines empty-set conventions explicitly and prevents duplicate predictions from inflating multiset matches. BOM metrics ignore poses, while edge F1 requires matched endpoints and the correct number of mating studs. Occupancy IoU is computed both without color and on cell/color tuples.

For an ordinary visual case, let  $S^+(T)$  be the set of colored first-hit cells along the positive X, Y, and Z axes. The visible structural predicate is

$$\begin{aligned} V(A, T) = & \mathbf{1}\{\text{valid}(A)\} \\ & \cdot \mathbf{1}\{\text{BOM}(A) = \text{BOM}(T)\} \\ & \cdot \mathbf{1}\{S^+(A) = S^+(T)\}. \end{aligned} \quad (5)$$

This is a nominal grid-surface proxy, not pixel similarity or a ray-tracing proof of what is visible from every perspective. Full-structure success is reported separately. A high surface F1 alone does not imply success.

Completion and editing additionally measure preservation of the protected source multiset. Completion reports precision/recall/F1 for additions after subtracting retained source matches. These addition diagnostics can disagree with a valid alternative hidden layout and therefore do not override ordinary-condition acceptance. Repair additionally requires exact faulty-current-ID localization, reports restoration and localization metrics separately, and measures false alarms only on normal cases.

Part recognition reports type, color, and stud-count accuracy. Relations report direct connectivity, vertical ordering, contact-stud count, and shortest undirected contact-graph distance. Constrained generation is scored against its six public constraints and legality; similarity to one arbitrary witness is not a generation-quality metric.

### 5.2. Executable Plans

A plan is evaluated in its submitted order. Execution stops at the first unknown, duplicate, unsupported, colliding, or blocked action. The scorer records the zero-based failure step, legal-prefix fraction, executed coverage, unique submitted coverage, length accuracy, and duplicate-free status.

Complete success requires every required ID exactly once and a fully executable sequence.

The evaluator does not sort a failed plan into a successful one. An incomplete but locally legal prefix is distinct from an attempted illegal action: its failure-step field is null, while success is zero. This distinction is important for greedy algorithms that run out of legal choices after prematurely covering a lower piece’s insertion path.

### 5.3. Denominators and Aggregation

A selection manifest freezes case IDs, filters, and a hash before scoring. Unknown or duplicate prediction IDs are rejected. Missing submissions produce explicit failure rows and remain in all applicable denominators. Malformed structure outputs receive zero structural metrics; malformed QA can retain correct-field partial credit but cannot receive exact success. A null metric means task-inapplicable, such as a false-alarm rate on a positive repair case, not an omitted failure.

The frozen v2 decoder had a JavaScript coercion gap: singleton arrays used as part-type or color fields could be accepted as string keys. New submissions use an explicitly versioned strict primitive-type gate; the legacy scorer remains available only for historical reproduction. All 117,910 oracle answers still pass, 5,120 malformed examples are rejected, and metrics on 118,640 existing baseline/API records are unchanged. Historical runs are not silently rewritten.

Reports are stratified by task, variant, condition, and split, with additional policy and piece-count summaries. Every metric carries its actual denominator. Success uncertainty uses a fixed-seed, 1,000-resample bootstrap of source-group means. No aggregate mixes ordinary images, layer disclosure, and full symbolic references. These intervals quantify sampling variation within this procedural resource, not semantic coverage or immunity to contamination.

## 6. Audit and Replay System

The replay format separates four actors: model, harness, environment, and evaluator. Events identify whether evidence was recorded or derived, and link content-addressed before/after states. Recorded API responses retain their original text and billing receipts. Decoding and deterministic reconstruction are marked as system operations. No unavailable internal reasoning or historical timestamp is invented.

For a structure collection, the viewer unfolds submitted parts as a harness visualization, not as an assembly plan. For a plan, it executes the original order and displays the first rejection. Input, intermediate output, and reference views can be compared with synchronized cameras. The input-only view hides the judge scenes and diagnostics. Per-case JSONL export preserves events and manifests.

| Direction   | Split      | Cases | Successes | %     |
|-------------|------------|-------|-----------|-------|
| assemble    | train      | 2,690 | 1,615     | 60.0  |
| assemble    | validation | 392   | 230       | 58.7  |
| assemble    | test_id    | 758   | 468       | 61.7  |
| assemble    | test_ood   | 1,280 | 51        | 4.0   |
| disassemble | train      | 2,690 | 2,690     | 100.0 |
| disassemble | validation | 392   | 392       | 100.0 |
| disassemble | test_id    | 758   | 758       | 100.0 |
| disassemble | test_ood   | 1,280 | 1,280     | 100.0 |

Table 2. Public-input greedy plan baseline on v2. The stored order is never repaired by the scorer. These are algorithmic, not neural, results.

The v2 external-submission track stores submitted JSON, not a fabricated provider transcript. Missing token counts, latency, and receipts are explicitly unavailable; zero new calls during replay is not a claim that an external model’s inference was free. Deterministic baseline outputs are labeled as harness/algorithm evidence rather than model responses. Every one of the 117,910 stored baseline cases has been replayed offline with score and state-chain checks; browser interaction tests are representative rather than 117,910 manual UI inspections.

## 7. Experiments and Current Results

### 7.1. Complete V2 Algorithmic Baseline

The baseline accepts only public task input. It computes symbolic relations, applies declared edit operations, greedily chooses an immediately legal next plan action, and copies or compares an explicitly supplied symbolic reference. It has no visual perception or constrained generation solver. For unsupported inputs it abstains; the evaluation records a zero rather than silently removing those cases. These abstentions are coverage facts, not failed attempts by a visual model.

All 117,910 selected cases were processed. The baseline solves all 10,240 relation cases, all 15,360 edit cases, and all 25,600 privileged symbolic reconstruction/completion/repair cases. It solves 2,364 of 5,120 assembly plans (46.2%) and all 5,120 disassembly plans. Table 2 separates the splits. Since this baseline is not learned, the ID/policy-held-out contrast describes procedural difficulty for a fixed algorithm, not learned OOD generalization.

The assembly/disassembly asymmetry is mechanistically interpretable. Current insertion legality does not ensure that a later lower piece will retain an unobstructed vertical corridor. A greedy strategy can place a supported roof or bridge early and block that corridor. Disassembly starts from the complete valid structure and removes an eligible piece. The result shows that our process metric distinguishes final geometry from executable construction; it



| Task/variant         | GPT-4.1 mini | Gemini Flash |
|----------------------|--------------|--------------|
| parts                | 8/25         | 12/25        |
| relations/contact    | 7/24         | 7/24         |
| relations/separated  | 0/24         | 0/24         |
| reconstruct/full     | 0/24         | 0/24         |
| generate/constraints | 0/24         | 0/24         |
| complete/suffix      | 0/24         | 0/24         |
| edit/rotate          | 0/24         | 0/24         |
| edit/recolor         | 24/24        | 24/24        |
| edit/remove          | 24/24        | 24/24        |
| plan/assemble        | 1/24         | 12/24        |
| plan/disassemble     | 1/24         | 7/24         |
| repair/none          | 21/24        | 17/24        |
| repair/color         | 4/24         | 6/24         |
| repair/shift         | 0/24         | 0/24         |

Table 3. New v2 one-shot results. Each structural variant has the same 24 source objects; the 25 isolated-part questions are separate.

does not show that the tasks are hard for a suitable search or dependency-aware planner.

A bottom-up height order, reversed for disassembly, solves all 10,240 plans and passes an independent execution checker. Under the current valid-grid and vertical-insertion assumptions, higher parts cannot block earlier lower insertions. This simple complete construction limits the planning claim to constraint following. A train-only BOM/dimension retrieval baseline was also evaluated on all 2,430 non-training objects for ordinary reconstruction; it neither sees target geometry nor adapts the retrieved structure to it.

## 7.2. V2 Fixed-Input Model Study

We freeze three held-out objects per policy, covering available size strata, before observing results: 24 objects, all 13 default/ordinary task variants, and 25 catalog questions yield 337 cases per model. GPT-4.1 mini and Gemini 2.5 Flash run interleaved at temperature zero with a 2,200-token output cap. All 674 calls, failures, raw responses, images, token counts, latencies, and settled receipts are retained. Table 3 gives exact successes; partial metrics and source-group bootstrap intervals are included in the artifacts.

For Gemini, one preselected object per policy receives six reconstruction controls and one assembly representation control (56 new calls). Reflection and legality feedback reuse the identical recorded first answer, charging only the new request. Feedback is target-blind. Table 4 does not support a claim that images are ignored: zero exact successes create a floor effect. Symbolic references provide additional privileged information; relative-coordinate plans use the same target information, not BrickNet connector programs.

Cross-task associations are computed over 24 source ob-

| Control            | Original | Control |
|--------------------|----------|---------|
| no-image           | 0/8      | 0/8     |
| mismatched-images  | 0/8      | 0/8     |
| symbolic-reference | 0/8      | 8/8     |
| reflection         | 0/8      | 0/8     |
| validation         | 0/8      | 0/8     |
| repeat             | 0/8      | 0/8     |
| relative           | 6/8      | 0/8     |

Table 4. Paired Gemini controls on eight objects. Relative denotes assembly; all other arms denote reconstruction.

jects, not 337 independent trials. Constant marginals yield undefined correlations. These are exploratory diagnostics without confirmatory power or a causal claim about reasoning or transfer.

## 7.3. Image-Conditioned Training Protocol

The new local pilot uses the public SmolVLM-256M-Instruct checkpoint [3] at revision 7e3e67ed, unquantized. Ninety-six training objects and 24 disjoint validation objects provide seven task families (672/168 examples) with actual supplied PNG views. Language-side query/value rank-8 LoRA has 460,800 trainable parameters. Single-reconstruction, multi-task, and leave-edit conditions each use seeds 17, 29, and 43; base is untrained. Every trained condition uses 120 updates, exactly 512 supervised answer tokens per update (61,440 total), AdamW at  $2 \times 10^{-4}$ , and gradient clipping at 1. Full prompts and answers are retained; loss-bearing answer positions are deterministically subsampled to match supervision.

This matches supervision, not input tokens, image tokens, time, or compute. Validation NLL uses two fixed examples per task, not all validation examples. Training uses PyTorch/MPS; every formal checkpoint uses MLX for greedy 2,200-token inference on the same 337 cases. Token, pixel, and mask digests match across both preprocessing backends on all cases; floating-point kernels need not produce identical output. All 840 training/validation prompt prefixes retain token identity when answers are appended; the longest full sequence is 5,176 tokens. Compact test-oracle answers use at most 1,793 SmolVLM tokens, below the 2,200-token cap. These private length diagnostics are never model inputs. Only completed jobs appear in Table 5; the artifact manifest records completed versus planned jobs. An incomplete matrix does not establish multi-task transfer.

## 7.4. Earlier Neural Pilots: Separate Evidence

Two preceding releases provide neural pipeline evidence but not a v2 leaderboard. A 576-object composition pilot evaluated four held-out objects with GPT-4.1 mini and Gemini 2.5 Flash. Each model answered 56 cases, for 112 API requests. Table 6 shows visual task counts; all

| Condition | Seed | Steps | Supervision | Success |
|-----------|------|-------|-------------|---------|
| base      | 17   | 0     | 0           | 0/337   |
| single    | 17   | 120   | 61,440      | 0/337   |
| single    | 29   | 120   | 61,440      | 0/337   |
| single    | 43   | 120   | 61,440      | 0/337   |

Table 5. Completed v2 local VLM jobs only. Each row evaluates 337 cases. All raw predictions are retained, including malformed responses.

| Task        | Model        | RGB | Layers | Symbolic |
|-------------|--------------|-----|--------|----------|
| reconstruct | GPT-4.1 mini | 0/4 | 0/4    | 4/4      |
| reconstruct | Gemini Flash | 0/4 | 0/4    | 4/4      |
| complete    | GPT-4.1 mini | 0/4 | 0/4    | 4/4      |
| complete    | Gemini Flash | 0/4 | 0/4    | 4/4      |
| repair      | GPT-4.1 mini | 0/4 | 1/4    | 3/4      |
| repair      | Gemini Flash | 2/4 | 0/4    | 3/4      |

Table 6. Earlier composition-v1 pilot only. Gemini Flash denotes Gemini 2.5 Flash. These 112 API calls are not v2 model evaluation.

cells have only four objects. Both models failed ordinary and layer reconstruction/completion, but succeeded with the privileged symbolic reference. This suggests a useful perception-versus-structure diagnostic, not a statistically established model comparison. V2 also changes some scoring rules, including full-structure requirements under layer disclosure; cross-version scores must not be pooled.

A further Gemini control reuses the identical first answer and input for 12 ordinary-condition generation/reconstruction/repair cases. One additional call either requests reflection without new scene information or supplies target-blind legality feedback. Both arms remain at 2/12 exact successes, with no gained or lost successes. The 24 new requests are billed once; reused historical receipts are not charged again. Arms ran sequentially rather than interleaved, and the sample is too small to establish that validation tools are generally ineffective.

An older Grid-v1 text/structure pilot uses Qwen3-0.6B and rank-8 LoRA on query/value projections, with 1,146,880 trainable parameters, AdamW learning rate  $2 \times 10^{-4}$ , and 120 optimizer steps per trained run. The maximum sequence length is 3,072; overlength training examples are excluded rather than truncating answers. Greedy inference allows 1,200 new tokens. Single-task training uses editing; multi-task training uses relations, generation, editing, and planning. Table 7 reports actual outputs, not a proposed recipe. All successes are editing cases. Multi-task seeds average 16.7% with a sample standard deviation of 7.2 percentage points, without evidence of broad transfer.

The single/multi runs are step-matched, not token-

| Condition | Seed | Steps | Tokens  | Success |
|-----------|------|-------|---------|---------|
| base      | 17   | 0     | 0       | 0/8     |
| single    | 17   | 120   | 195,292 | 2/8     |
| multi     | 17   | 120   | 173,957 | 1/8     |
| multi     | 29   | 120   | 176,765 | 1/8     |
| multi     | 43   | 120   | 173,137 | 2/8     |

Table 7. Earlier Grid-v1 local text/structure pilot, eight test cases per row. Tokens count processed input plus output training tokens. Base/single have one seed each. This is not v2 or multimodal training.

matched, and use different supervision diversity. A lower validation NLL is not evidence of execution success. Four adapters, five prediction sets, and all 480 optimizer-step records are bundled. Training used an Apple M4 Pro with 48 GB memory and MPS; local computation consumed real resources despite having no API bill.

## 7.5. Verification and Resource Accounting

All 117,910 v2 oracle answers pass and all empty answers fail. The artifact verifier regenerates every ground truth and compares all public inputs and private render recipes, checking 476,760 serialized rows across 14 compressed shards. Ground truth is therefore materialized and checked case by case, rather than inferred from a few examples. A full offline replay independently re-evaluates the deterministic baseline and reconstructs all event state chains.

Engineering validation passes 85 benchmark tests and 69 original application unit tests, TypeScript checking, and production frontend build. Browser checks cover 1,440-pixel desktop and 390-pixel mobile layouts, nonblank canvas pixels, camera interaction, input/truth separation, pagination, scoring, and replay export. A render audit tests 115 views across the largest sample in each policy. All dataset render recipes are verified, but not all possible PNGs are pre-rendered. A 46-case external-input export verifies 125 image uses and 25 unique image hashes.

For the new study, all 1,177 train/validation/test input records are checked against their task recipes: 1,970 image uses and 626 distinct renders match by regeneration. The portable training bundle contains 840 records and 480 unique PNGs with verified input/answer and split identity. Rendering equality is checked on the same browser/GPU stack, not claimed across arbitrary rasterizers.

A copied environment without the original runtime directory or API key regenerates the complete casebank audit and replays all evaluations. It shares the installed pinned Node dependencies; this is not a clean operating-system installation, a second-person reproduction, or a repeated stochastic model training run.

The earlier API campaign contains 363 settled requests totaling \$0.45102511: \$0.10992152 for CARE-mini

and diagnostics, \$0.08908686 for Grid-v1 experiments, \$0.20204040 for the composition pilot, and \$0.04997633 for the paired controls. The new v2 study adds 730 settled requests costing \$1.50635072, bringing the cumulative campaign to 1,093 requests and \$1.95737583. Construction and offline verification add no API requests. We preserve the cumulative ledger and its \$4.50 internal cap rather than resetting the budget between releases.

## 8. Limitations and Research Readiness

**Data and domain.** The casebank contains procedural, not human-authored, structures and shares a common root plate. More task conditions do not create more independent objects. Eight policies do not imply broad semantic coverage. The known-part recognition set is not a novel-part generalization test. Policy-held-out bridge and shell data share primitives with training. Near-duplicate checking is exhaustive only for its declared token representation.

**Validity of ground truth.** Two geometry implementations reduce some coding risks but share the same nominal assumptions. No force, clutch, deformation, load-bearing stability, or arbitrary CAD collision model is validated. Public BrickNet connector annotations were imported in an earlier audit, but only the 25-type grid domain was cross-checked; the gated human-model dataset was not obtained. Real independent human review is still absent. The 200-object review interface does not constitute 200 completed reviews.

**Metric validity.** The surface criterion reduces hidden-layout unfairness but is not a complete model of image observability. It ignores some rendered seam information and can differ from perspective visibility. Exact metrics can be unforgiving to near-correct answers; partial metrics are included to make that distinction visible. Additional calibration using independent human judgments and adversarial alternatives is needed.

**Empirical scope.** The new v2 neural study covers only 24 shared structural objects and two API models, plus a small local VLM training protocol. Earlier studies use different data and remain separate. Exploratory correlations and eight-object controls cannot establish general representation advantages or causal multi-task transfer. Broader model families, specialized baselines, semantic diversity, and confirmatory sample designs remain necessary before strong comparative claims.

**Release and research ethics.** Procedural model records declare CC0 provenance. Imported BrickNet annotations retain MIT attribution, and the Qwen base checkpoint and

adapter documentation retain the Apache-2.0 context. Raw API receipts and outputs are preserved without storing credentials or machine identity. A future human study needs actual independent participants, appropriate consent, and a documented disagreement process. This manuscript uses the official CVPR 2026 template for exposition; it is not a claim of submission, acceptance, or compliance with a future conference year’s rules.

## 9. Conclusion

BrickAtlas v2 turns a small multi-task prototype into a materialized, versioned casebank with per-case ground truth, explicit information conditions, complete metric denominators, and auditable replay. Its full algorithmic audit reveals both greedy failure and a simple complete height-sorted planner, preventing an overstated search claim. The v2 model study exposes failures in visual reconstruction alongside stronger symbolic editing, while preserving original outputs and costs. Procedural diversity, nominal geometry, and small-sample model evidence remain no substitutes for independent data review and broad evaluation.

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